Elderly Care Management System

Low-Fidelity Paper Wireframe & Prototype Design Specification

STUDENT DETAILS

Name: Farzana Sadia Takee
Student ID: 2211175
Course Project: Elderly Care Management System
Submission Type: Paper Wireframe / Low-Fidelity Design

DESIGN PHILOSOPHY & ACCESSIBILITY SPEC

- **Senior Accessibility:** High contrast, ≥56px touch targets, and **14pt (18.67px) minimum body text** in Senior Resident UI (14px minimum across clinical tables).
- **5-Role Portal Architecture:** Dedicated screens for Family, Caregiver, Senior, Physician, and Administrator.
- **Safety First:** Instant 1-touch 999 SOS countdown, eMAR dose tracking, and HIPAA audit logs.

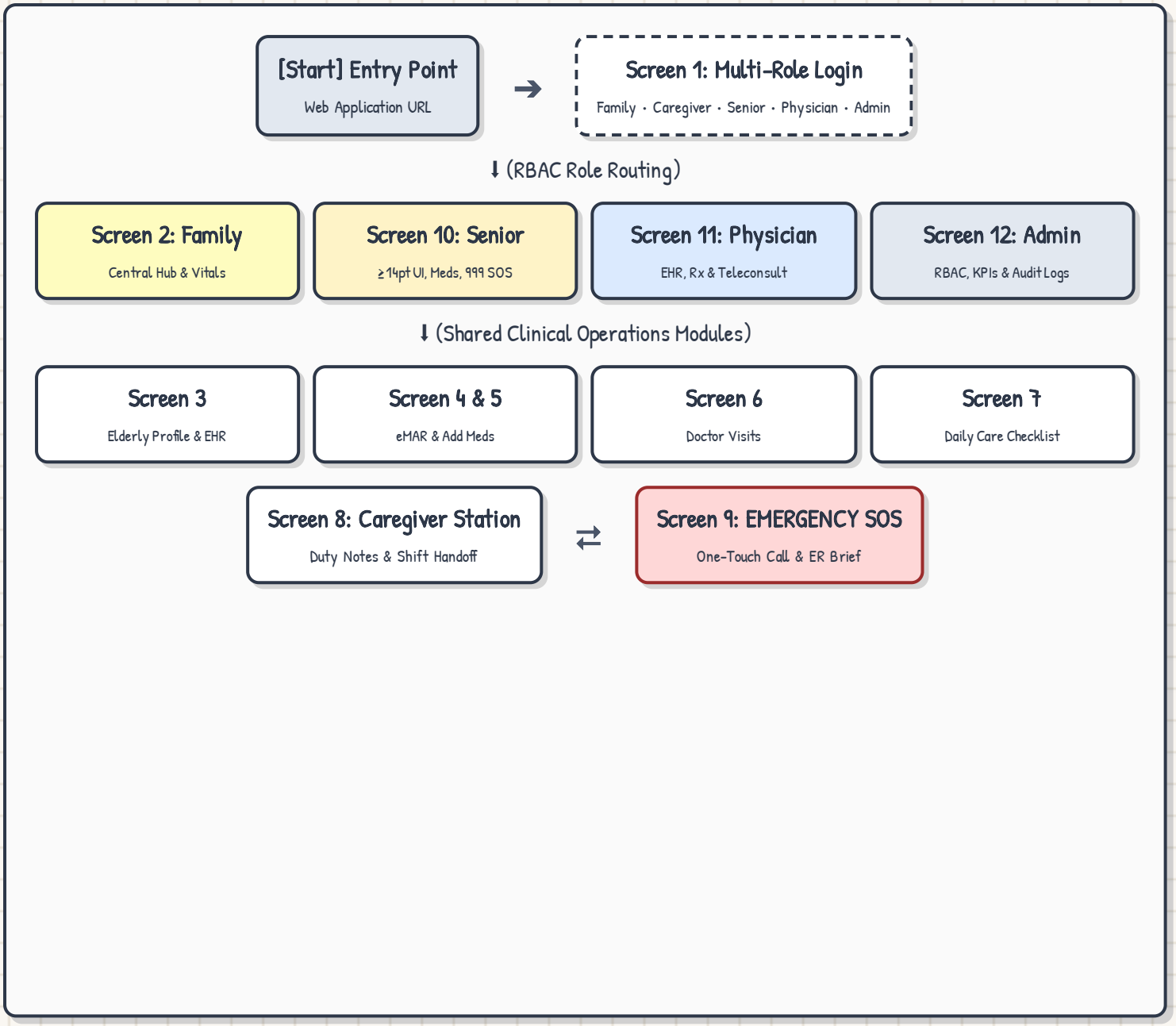
SEQUENCED INTERFACE INDEX (14 PAGES • 12 SCREENS)

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3. Screen 2: Family Member Dashboard
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01. Navigation Flow & 5-Role System Architecture

Role-based routing from authentication to dedicated Senior, Physician, Admin, Family, and Caregiver portals

Sequence Map



User Flow Rationale

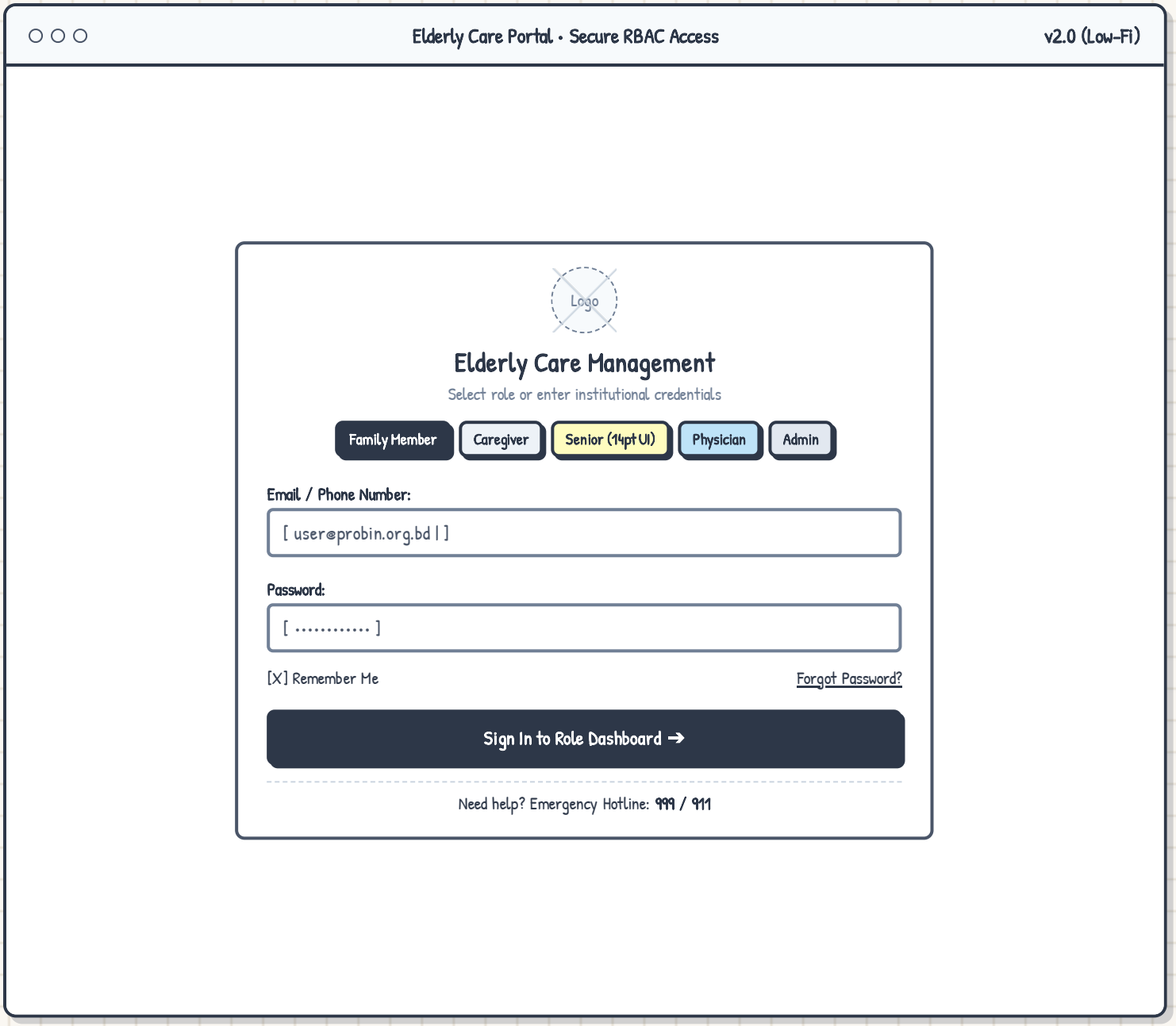
- **5-Role Isolation:** Each user lands only on tools authorized for their clinical or personal role.
- **Reconciled Accessibility:** Senior portal enforces ≥14pt (18.67px) text & giant buttons.
- **Persistent SOS:** Emergency trigger reachable across portals.

5 Target Personas

- **Family & Caregiver:** Remote monitoring & daily shift care.
- **Senior Resident:** Oversized meds, dial & 999 SOS UI.
- **Physician & Admin:** Clinical Rx/EHR & RBAC audit governance.

02. Screen 1: Landing & 5-Role RBAC Login

Clean authentication portal with 5-role selector (Family, Caregiver, Senior, Physician, Admin)



UI Wireframe Notes

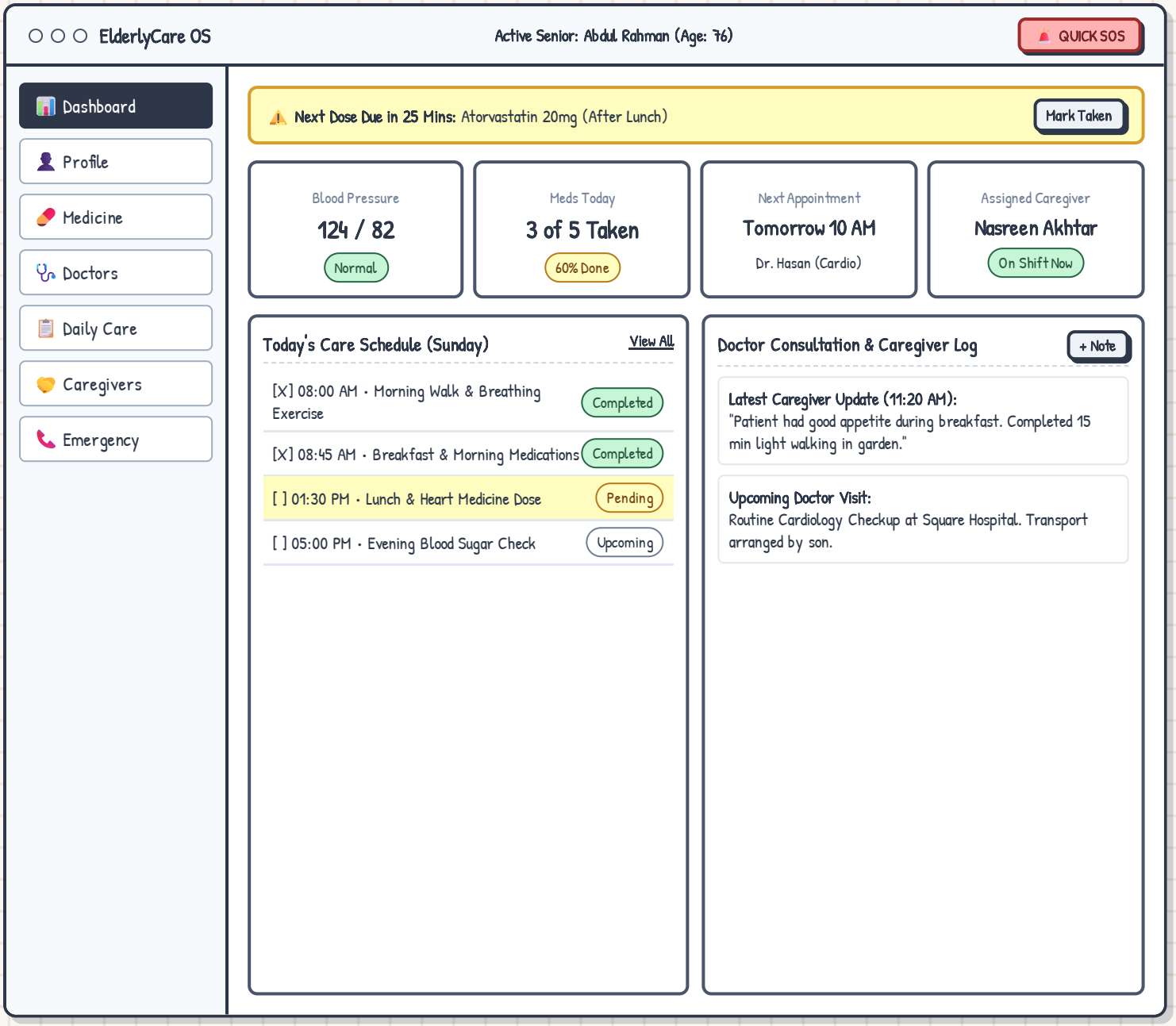
- **5-Role Toggle:** Routes users directly to Family (S2), Caregiver (S8), Senior (S10), Physician (S11), or Admin (S12).
- **High Accessibility:** Large input boxes and ≥48px controls.
- **Quick Recovery:** Emergency helpline visible before login.

User Actions

- Select persona or enter JWT email/password.
- Auto-route to role-specific dashboard.

03. Screen 2: Family Member Central Dashboard

High-level overview: active patient vitals, upcoming meds, care schedule, and rapid SOS



Dashboard UX Highlights

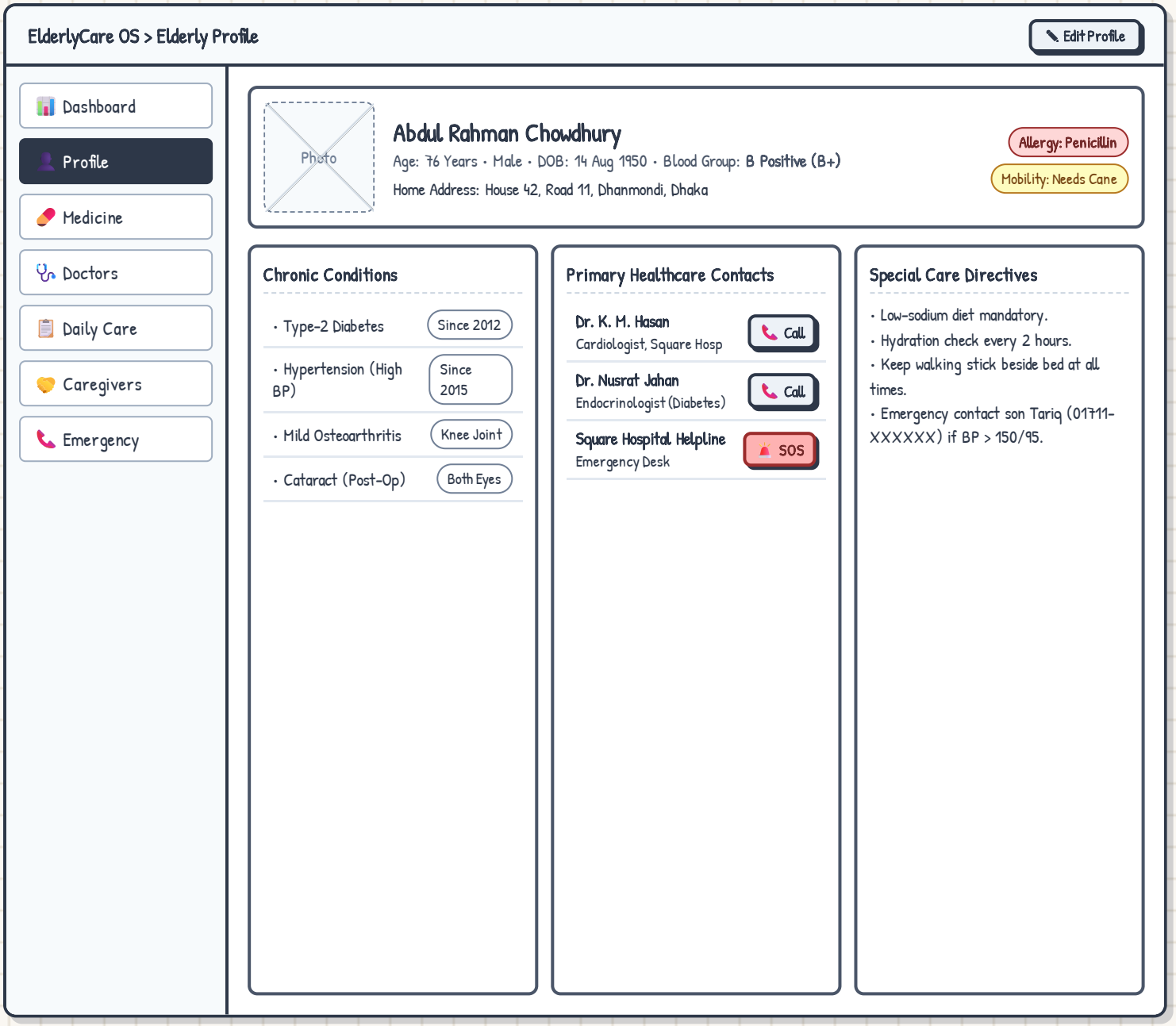
- **One-Glance Digest:** Family members immediately see if any medication is missed.
- **Vitals Cards:** Shows latest BP, blood sugar, and caregiver shift status.
- **Red SOS Button:** Positioned in header on every page for extreme emergencies.

Key Interactions

- Click 'Mark Taken' to record medication adherence.
- Click any nav item to jump to dedicated module.

04. Screen 3: Elderly Profile & Medical History

Comprehensive profile: personal data, chronic illnesses, blood group, allergies, and doctors



Design Considerations

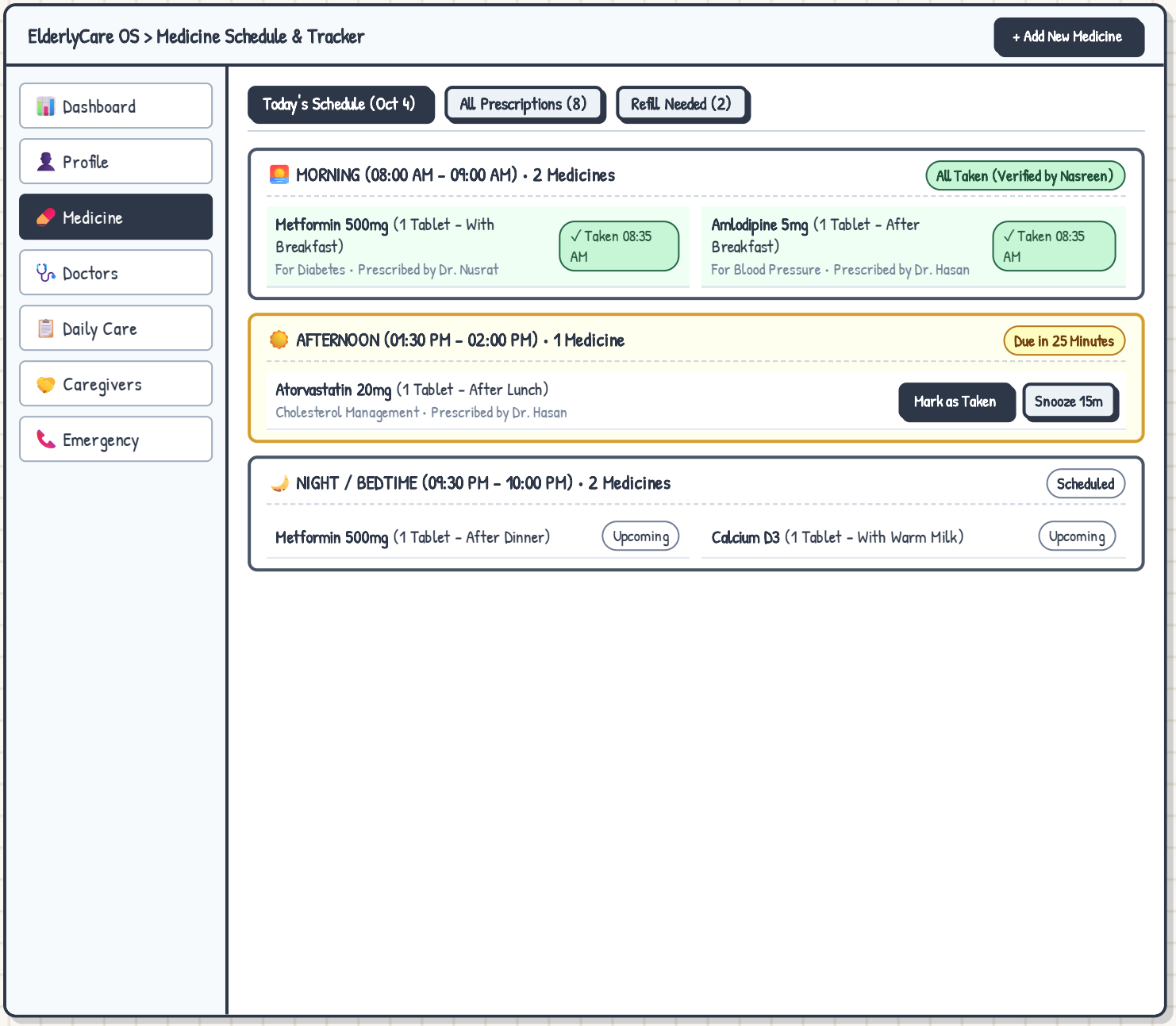
- **Critical Health Flags:** Blood group and Penicillin allergy stand out prominently.
- **Care Directives:** Clear instructions for newly arriving or rotating caregivers.
- **Quick Call Actions:** Fast dial buttons for primary treating physicians.

Data Handled

- Demographic & address details
- Medical conditions history
- Physician association records

05. Screen 4: Medicine Schedule & Reminders

Daily medication timeline categorized by Morning, Afternoon, Evening, and Bedtime



Medicine Reminder Logic

- **Chronological Blocks:** Morning, Afternoon, Evening, Bedtime groupings avoid confusion.
- **Actionable Buttons:** Simple one-click "Mark as Taken" or "Snooze".
- **Audit Trail:** Displays exact timestamp and caregiver name who verified dosage.

Safety Impact

- Prevents double-dosing and forgotten medicines.
- Alerts family instantly if dose is overdue by > 30 mins.

06. Screen 5: Add / Edit Medication Form

Modal / form for family members or doctors to input medication dosage, frequency, and instructions

ElderlyCare OS > Add New Prescription [X] Close Window

Add Medication Schedule

* Required Fields

Medicine Name:*
[Atorvastatin Lipitor I]

Dosage & Unit:*
[20 mg (1 Tablet) v]

Frequency:*
[Once Daily v]

Timing Routine:*
[After Lunch v]

Time of Day:*
[01:30 PM]

Prescribed By Doctor:
[Dr. K. M. Hasan (Cardiology) v]

Current Stock / Refill Alert:
[30 Pills (Alert at 5)]

Special Instructions & Caution:
[Do not take with grapefruit juice. Take with full glass of water.]

[X] Enable SMS & App push reminder alerts

Cancel | Save & Activate Schedule

Form Field Validation

- **Dosage Safeguards:** Dropdowns ensure standard medical units (mg, ml, drops, puffs).
- **Instruction Box:** Prevents dietary contraindications (e.g. food requirements).
- **Refill Automation:** Tracks remaining pills and triggers re-order alert.

User Inputs

- Drug name and formulation
- Exact intake time & meal relation
- Notification preferences

07. Screen 6: Doctor Appointment Tracking

Healthcare appointments calendar, consultation logs, and medical booking manager

ElderlyCare OS > Doctor Appointments

+ Book / Schedule Visit

- Dashboard
- Profile
- Medicine
- Doctors
- Daily Care
- Caregivers
- Emergency

NEXT UPCOMING APPOINTMENT

Dr. K. M. Hasan — Senior Cardiologist

Square Hospital, 3rd Floor, Room 314 · Tomorrow (Monday, Oct 5) at 10:30 AM

Purpose: 3-Month Echo & Blood Pressure Review · Accompanied by: Tariq (Son)

View Clinic Map

Reminder Active

Scheduled Future Visits

Dr. Nusrat Jahan (Endocrinologist)
Labaid Diagnostic, Dhanmondi
Oct 18, 2026 · 05:00 PM · Fasting Glucose Test
Reschedule

Dr. Shahed Ali (Orthopedic)
Popular Hospital · Knee Pain Checkup
Nov 02, 2026 · 11:00 AM
Reschedule

Past Consultation History & Prescription Reports

Routine Health Checkup — Sep 04, 2026
Dr. K. M. Hasan · BP adjusted to 120/80
View RX

Ophthalmology Eye Exam — Aug 12, 2026
Vision stable, no laser required
View RX

Appointment UX Highlights

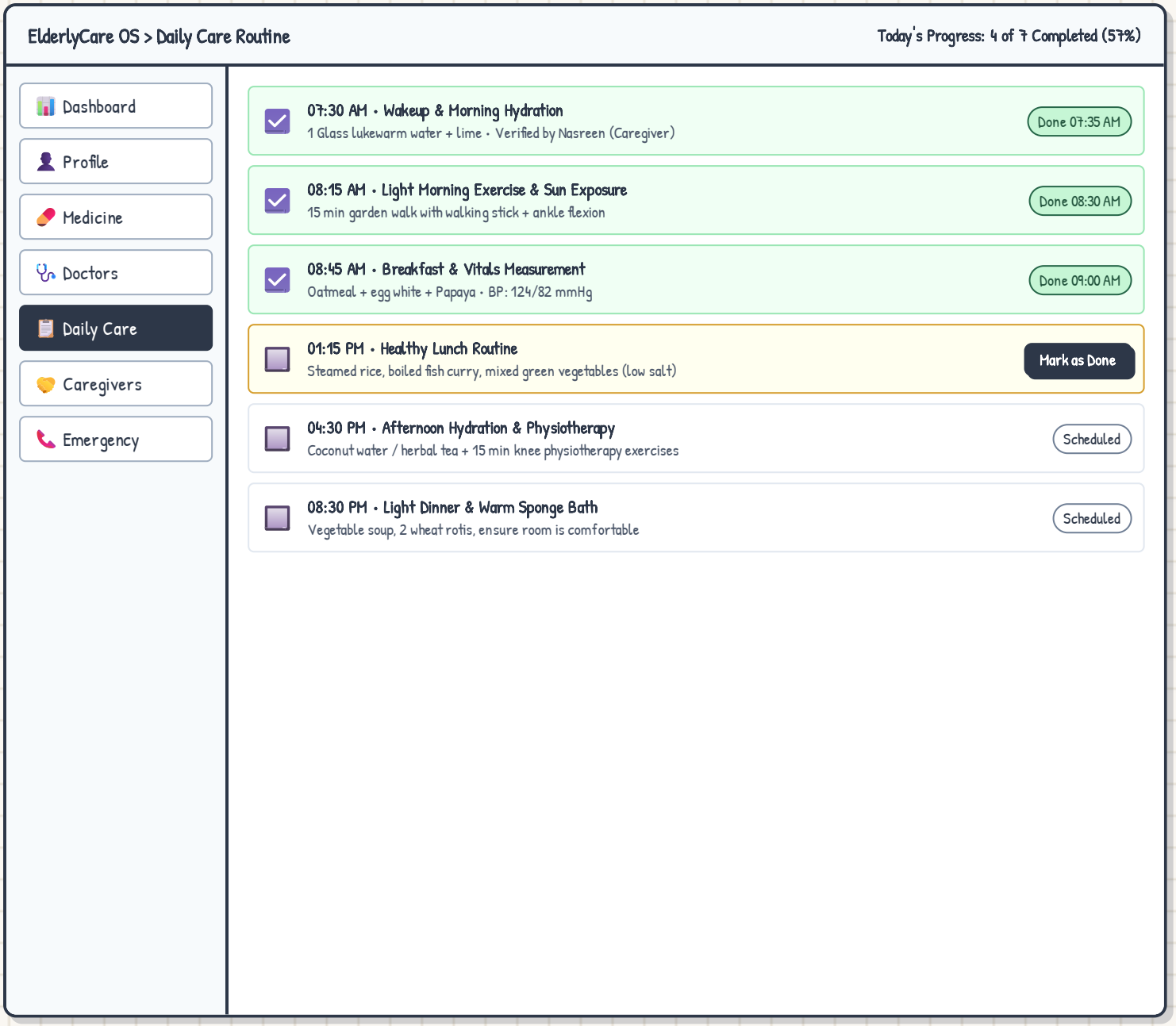
- **Hero Alert Card:** Immediate visibility of nearest upcoming doctor appointment.
- **Accompaniment Info:** Identifies which family member will escort the senior.
- **Prescription Archive:** Quick link to download previous doctor notes & prescriptions.

Reduces Missed Visits

- Automated 24h and 2h SMS reminders to both caregiver and family member.

08. Screen 7: Daily Care Schedule & Activities

Routine checklist for meals, hygiene, physical mobility, hydration, and vitals recording



Holistic Daily Care

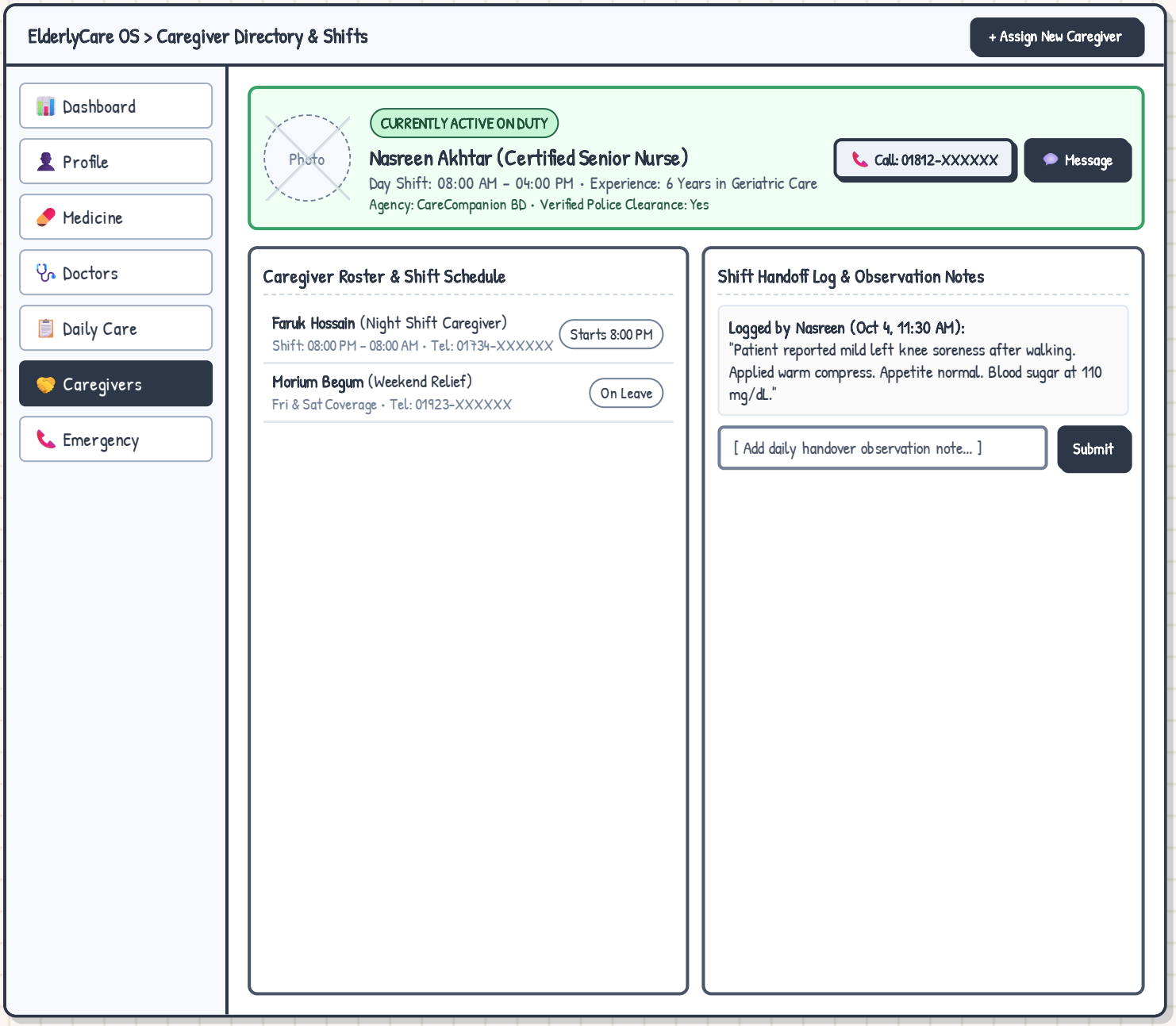
- **Beyond Medicine:** Incorporates hydration, diet, exercise, and mental stimulation.
- **Caregiver Handoff:** Checked items generate a daily summary report for family.
- **Large Checkboxes:** Easily tapped on mobile devices by attending staff.

Task Types

- Meal & nutrition routines
- Physical therapy / mobility
- Hygiene & vitals logging

09. Screen 8: Caregiver Information & Shift Handoff

Caregiver roster, assigned shifts, duty checklists, direct calling, and daily observation logs



Caregiver Management UX

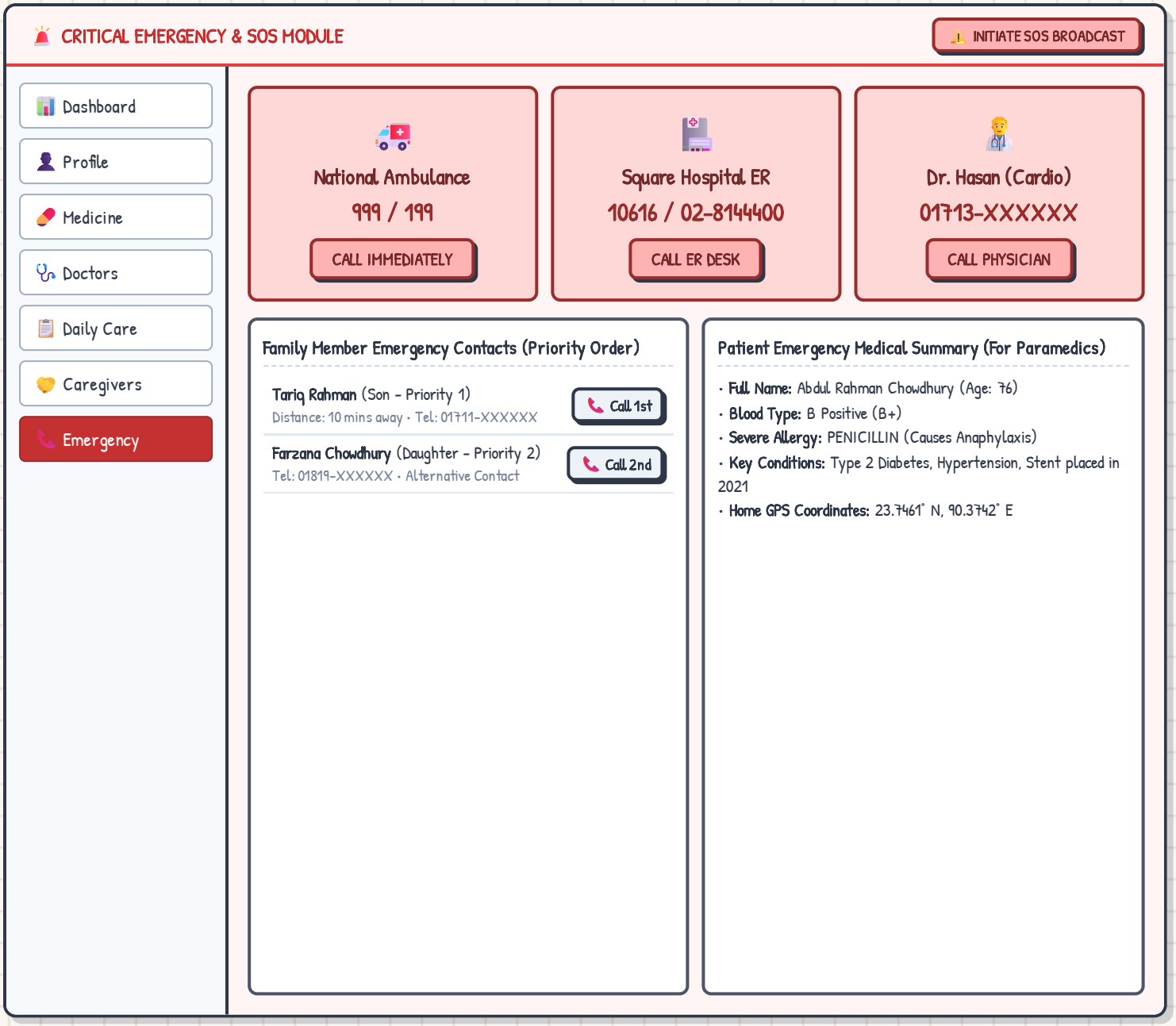
- **Direct Connectivity:** Family can phone or text the caregiver on duty with 1 tap.
- **Digital Shift Handover:** Prevents critical care omissions when shifts change.
- **Vetted Records:** Shows certification and agency contact info for transparency.

Accountability

- Shift clock-in and check-out logs.
- Mandatory shift-end handover remarks.

10. Screen 9: Emergency Contacts & Quick SOS Protocol

High-priority rapid response portal with single-tap emergency calling and hospital routing



Emergency UI Principles

- **High Affordance:** Massive, non-confusing call buttons.
- **Instant Paramedic Brief:** Blood type and Penicillin allergy instantly available to arriving medics.
- **One-Tap SMS Broadcast:** Notifies all family members simultaneously with GPS pin.

Compliance

- Meets ISO medical emergency alert standards for senior citizens.

11. Screen 10: Senior Resident Accessible Dashboard

Zero-clutter senior portal with ≥14pt (18.67px) text floor, ≥56px buttons, med reminders & 999 SOS

ElderlyCare Senior Mode · No Sidebar / Distraction-Free

Min Body Text: 14pt (18.67px)

Hello, Alhaj Shamsuddin Sahib!

Suite 204 · Duty Nurse: **Fatema Begum (RN)**

WCAG AAA 7:1

HELP
SOS 999

Need Immediate Help? Tap Big HELP Button

5-sec audible countdown alerts Nurse & Son Tanvir.

[X] CANCEL GUARD

My Medicines Today

Tab. Comet 500mg
08:30 AM · Morning
[✓]

Tab. Camlosart 5/20
08:30 PM · Tap box
[ ]

Today's Routine

08:30 · Breakfast & BP
Served with meds

11:00 · Garden Walk
15-min gentle walk

03:30 · Video Call
Call Son Tanvir

1-Tap Speed Dial

Oversized buttons (≥56px):

Call Son Tanvir

Call Duty Nurse

Reconciled Typography

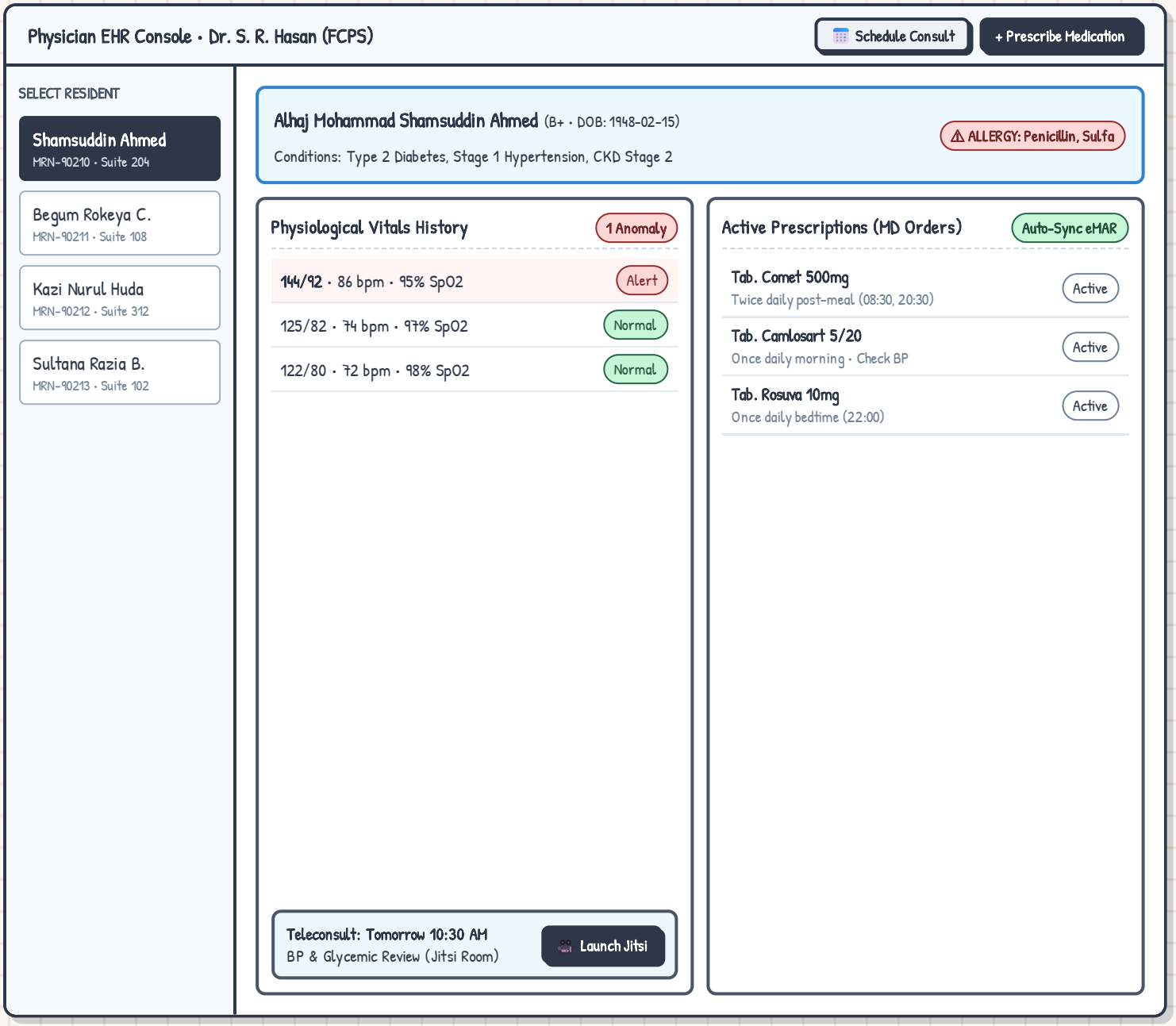
- **Strict 14pt (18.67px) Floor:** All senior reminders & buttons are ≥14pt, matching Page 1 & High-Fi Design System tokens.
- **Zero Sidebar Clutter:** Full-width layout removes tiny menu trees for elderly users.

Safety & Touch Targets

- **5-Sec Tap Guard:** Audible countdown prevents accidental 999 false alarms.
- **≥56px Buttons:** Easy tapping for arthritis or tremor sensitivity.

12. Screen 11: Physician Clinical EHR & Consultation Portal

Doctor workspace for medical records review, drug allergy checks, e-prescribing & video consultations



Physician Workflow UX

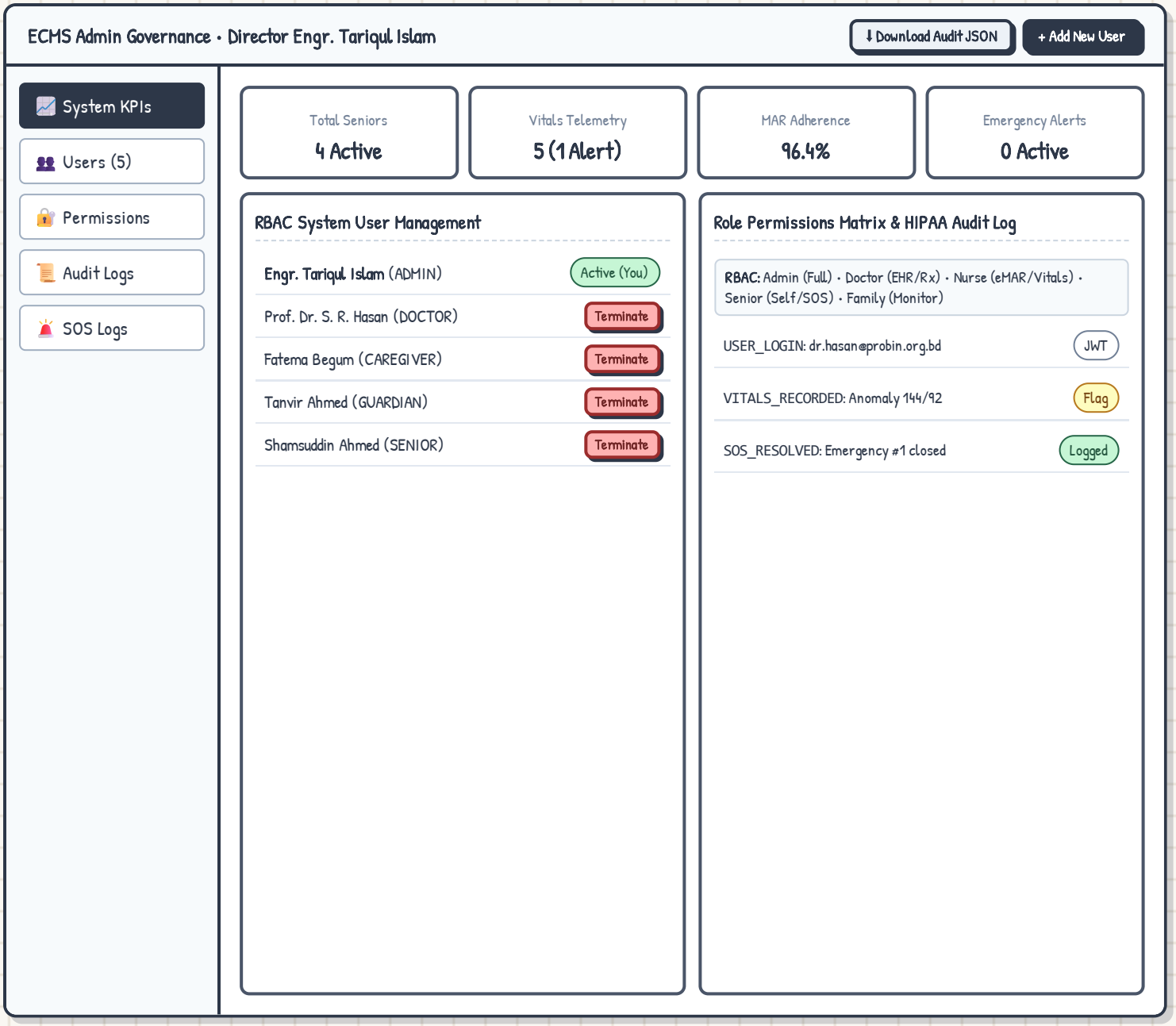
- **Allergy Guardrail:** High-contrast drug allergy tag pinned above prescription triggers.
- **Order-to-eMAR Sync:** New MD orders automatically create scheduled PENDING slots for duty nurses.

Clinical Separation

- Doctors diagnose, prescribe & consult via encrypted Jitsi rooms.
- Caregivers execute & sign off daily doses without prescribing authority.

13. Screen 12: Administrator Governance, RBAC & Audit Dashboard

System monitoring KPIs, user account management, role permissions matrix & HIPAA audit logs



Governance & RBAC

- **Lifecycle Provisioning:** Create accounts and terminate or reactivate access instantly.
- **Self-Lockout Guard:** Prevents admins from terminating their own active session.

Monitoring & Audits

- Tracks resident KPIs and anomaly counts live.
- Immutable timestamped HIPAA audit trail with JSON export.